The Liouville skeleton in the skew-symmetrizable acyclic case

Hypotheses, geometric target, and formalization scope

Cluster skeleton formalization project

Abstract

We specify the skeleton theorem for an integral unimodular completion E equipped with a positive rational diagonal matrix $\widetilde{D}$ such that $K = E\widetilde{D}$ is skew-symmetric. The accompanying blueprint gives the weighted geometric argument. The existing Lean library proves the special case $\widetilde{D} = I$; the general statement below is the current formalization target, not a claim of completed Lean verification.

1 Input data and exchange space

Let $n \geq 0$, let $M \subseteq \{1, \dots, n\}$ be mutable, and let F be its complement. Fix

$$E \in GL_n(\mathbb{Z}), \qquad \widetilde{D} = \operatorname{diag}(d_1, \dots, d_n), \quad d_j \in \mathbb{Q}_{>0}, \qquad K = E\widetilde{D}, \quad K^{\mathsf{T}} = -K.$$

Thus $E_{ij}d_j = -E_{ji}d_i$, rather than $E_{ij} = -E_{ji}$. Require the mutable block $B = E_{M,M}$ to be acyclic: the arrows $i \to j$ with $B_{ij} > 0$ have no directed cycle. No acyclicity is required on the full set of indices. The unimodular completion is part of the input; existence of such a completion for every exchange matrix is not asserted. In fact $\det E = 1$: a nonsingular real skew-symmetric matrix has positive determinant, so $\det E = \det K / \det \widetilde{D} > 0$, while unimodularity gives $\det E = \pm 1$.

Use *column* exchange exponents throughout:

$$P_i^{\pm}(z) = \prod_{j=1}^{n} z_j^{[\pm E_{ji}]_+}.$$

$$X_E = \left\{ (z, z') \in \mathbb{C}^n \times \mathbb{C}^M : \begin{array}{ll} z_i z_i' = P_i^+(z) + P_i^-(z) & (i \in M), \\ z_f \neq 0 & (f \in F) \end{array} \right\}.$$

Here $[a]_+ = \max(a, 0)$. Give X_E the induced topology. Frozen coordinates remain variable elements of $\mathbb{C}^*$; no principal-coefficient assumption is made. The initial torus is $T = \{z_1 \cdots z_n \neq 0\} \subset X_E$. On it set

$$c_i = Ee_i, \qquad t_i = z^{c_i} = P_i^+ / P_i^-, \qquad \Omega_T = \frac{1}{2} \sum_{j,l} K_{jl} \, \mathrm{d}\log z_j \wedge \mathrm{d}\log z_l.$$

The integer exponents come from E, not from the generally rational K.

2 Independently defined subsets

Define the coordinate cap without reference to any flow:

$$C_E = \{(z, z') \in X_E : |z_f| = 1 \ (f \in F), \quad |z_i| \le 1, \ (1 - |z_i|^2) z'_i = 0 \ (i \in M)\}.$$

Call $I \subseteq M$ independent if $E_{ij} = 0$ for all $i, j \in I$, and put

$$\Lambda_I = \{(z, z') \in X_E : z'_i = 0, \ |z_i| \le 1 \ (i \in I), \quad |z_j| = 1 \ (j \notin I)\}.$$

For a real flow Φ on X_E, independently define

$$\mathrm{Skel}(\Phi) = \{p \in X_E : \overline{\{\Phi_t(p) : t \ge 0\}}^{X_E} \text{ is compact}\}.$$

The closure is taken in the actual exchange space.

3 The weighted geometric target

Theorem 3.1 (Skew-symmetrizable skeleton target)**.** *For every $(n, E, \widetilde{D}, M)$ satisfying the hypotheses above, the fixed space X_E admits a smooth real manifold structure of dimension $2n$, a smooth real one-form λ, a smooth vector field Z, and a jointly smooth flow $\Phi : \mathbb{R} \times X_E \to X_E$ such that:*

1. *The actual coordinate inclusion in $\mathbb{C}^n \times \mathbb{C}^M$ is a smooth immersion, and T is dense.*
2. *$\omega = \mathrm{d}\lambda$ is nondegenerate and $\omega|_T = \mathrm{Im}\, \Omega_T$.*
3. *$\iota_Z \omega = \lambda$, $\Phi_0(p) = p$, $\Phi_{t+s}(p) = \Phi_t(\Phi_s(p))$, and $\frac{\mathrm{d}}{\mathrm{d}t}\Phi_t(p) = Z(\Phi_t(p))$ for every real s, t. Thus this same field is complete in both time directions.*
4. *For this same flow,*

$$\boxed{\mathrm{Skel}(\Phi) = C_E = \bigcup_{I \subseteq M \text{ independent}} \Lambda_I.}$$

   Each Λ_I in this union is compact.

For independent I, the node chart $U_I = \{z_j \ne 0 \ (j \notin I)\}$ has the form $\mathcal{N}^I \times (\mathbb{C}^*)^{n-2|I|}$, where $\mathcal{N} = \{(x, y, t) : xy = 1 + t, \ t \ne 0\}$; in particular $2|I| \le n$.

The quantifier order is $\forall (n, E, \widetilde{D}, M)$ $\exists$(smooth structure, λ, Z, Φ). No flow, primitive, completeness, nondegeneracy, or skeleton inclusion is supplied as an additional hypothesis. The real smooth theorem is the Lean objective; the blueprint also explains the local extension of Ω_T.

4 Why the weights cannot be discarded

Write $z_j = e^{u_j + \mathrm{i}\theta_j}$, $v_i = c_i(u)$ and $\eta_i = c_i(\theta)$. The blueprint constructs

$$\lambda_{\mathrm{tor}} = u^{\mathsf{T}} K \, \mathrm{d}\theta, \qquad H = \sum_{i \in M} d_i h(u_i) \chi_i \rho(v_i) \beta(\eta_i), \qquad \lambda = \lambda_{\mathrm{tor}} + \mathrm{d}H.$$

The identities $K e_i = d_i c_i$ and $c_i^{\mathsf{T}} K^{-1} = -e_i^{\mathsf{T}}/d_i$ give

$$\dot{u}_i = u_i - h(u_i) \chi_i \rho(v_i) \beta'(\eta_i), \qquad \dot{\eta}_i = d_i^{-1} \partial_{u_i} H.$$

The radial equations have the same form as in the unweighted case because the weights cancel, not because d_i is assumed to be one. Every node symplectic block and its primitive carries the factor d_i. Positivity of d_i and invertibility of K enter the construction.

A concrete input that the general theorem must accept is

$$E = \begin{pmatrix} 0 & 1 & -1 & 0 \\ -2 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \widetilde{D} = \operatorname{diag}(1,2,1,2), \qquad M = \{1,2\}.$$

Here $\det E = 1$, $E\widetilde{D}$ is skew-symmetric, and the mutable quiver is acyclic, but $E_{12} = 1 \neq 2 = -E_{21}$. This example checks that the input is genuinely broader; it does not replace the theorem quantified over all admissible inputs.

5 Lean correspondence and acceptance

The existing source is the starting library. Its current `Seed.skew` requires a skew-symmetric integer matrix, and `FullSkeleton.firstGeometricTarget` proves that specialization. Existing verification receipts therefore certify only that baseline. They must not be reported as verification of Theorem 3.1.

The generalized seed must distinguish integral exchange data E from the weighted form $K = E\widetilde{D}$. The existing modules for zero sets, independent charts, smooth forms, cutoffs, global primitives, complete flows and both skeleton inclusions must be adapted on the same actual exchange space. Reuse existing constructions and proofs wherever valid; replacing the entire geometry by an assumed witness is not a proof.

The old code uses row exchange exponents. For $\widetilde{D} = I$, row and column conventions only interchange the two monomials. For general $\widetilde{D}$, this argument fails: the magnitudes can differ. The implementation must use the column convention above, or prove an explicit transpose/convention equivalence preserving the exchange space, the sign of the form, and the same-flow skeleton identity.

Acceptance requires the quantified weighted theorem, recovery of the old specialization, the non-skew example above, full library and final consumer compilation, and a transitive axiom audit allowing only `propext`, `Classical.choice`, and `Quot.sound`. No `sorry`, added geometric axiom, modified dependency, or assumption equivalent to the desired theorem is allowed. Updated receipts must bind the new source and the generalized theorem types.

This target retains unimodularity and mutable acyclicity. It makes no claim for singular completions, arbitrary cyclic seeds, generalized cocores, or categorical mirror symmetry.